

SCOUT: An AI-Powered Vision-Based Fall Detection System for Elderly Safety

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Abstract: Falling represents one of the largest public health threats to the elderly, and falls have become a major issue for the world's healthcare systems. This research project introduces SCOUT, a home-based, continuous, computer-vision based fall detection system which uses a YOLOv8n deep learning model for the purpose of privacy preserving monitoring at home. The SCOUT system integrates a lightweight object detector with temporal heuristics, and an optional pose verification module, to provide both fast and accurate fall alerts, while providing strong user privacy protection by locally computing the object detector results and minimizing transmitted data. Because of its design for edge-based deployments, the SCOUT system also has scalable, affordable, and reliable operations in all resource limited environments. A publically available dataset was curated and expanded for training the SCOUT model, specifically for use in detecting falls in older adults. These experimental results show that SCOUT achieved the desired performance metrics of detection accuracy = 93%, precision = 0.94, recall = 0.945 and F1 score = 0.95, while demonstrating robustness across various lighting, occlusion, and camera angle challenges. The results shown above clearly support that SCOUT is capable of adapting to different use cases, and

therefore applicable to Smart Home and Elderly Care Monitoring applications.

Keywords: *Deep Learning, YOLOv8n, Neural Networks, Artificial Intelligence, and Computer Vision/ Image Processing*

I. INTRODUCTION

Falls by elderly people represent a major public health concern and have serious consequences such as injury, loss of independence, hospitalization, and a significant economic burden on healthcare systems. In order to mitigate these long term negative impacts, early detection and rapid response are critical. As non-intrusive vision-based monitoring does not require the person to be wearing a device continuously monitors them, this represents a promising solution for fall detection within the home environment. The SCOUT (Smart Camera Observations and Unusual Tracking) system is designed to be a camera-only system which provides rapid person detection, along with posture heuristics and optional pose verification, providing timely alerts with evidence clips for caregivers. Therefore, this document outlines all aspects of the design, experiments, evaluation and analysis needed for